```
#Dictionary ,key and value can be any data type, items =key/value pair
students = {"ali":45,"uzair":30,"bashir":25}
students
```

```
{'ali': 45, 'uzair': 30, 'bashir': 25}
```

```
students_1 = {"ali":[45,36,25],"uzair":[50,45,25],
              "bashir":[50,45,25]}
students_1
```

```
{'ali': [45, 36, 25], 'uzair': [50, 45, 25], 'bashir': [50, 45, 25]}
```

```
students["ali"]
```

```
45
```

```
students["ali"] = "aleena"
students
```

```
{'ali': 'aleena', 'uzair': 30, 'bashir': 25}
```

```
employee = {"name":"moin","father_name":"khan",
            "designation":"xyz","age":25}
employee["department"] = "logistic"
```

```
employee
```

```
{'name': 'moin',
 'father_name': 'khan',
 'designation': 'xyz',
 'age': 25,
 'department': 'logistic'}
```

```
#Methods
#remove it need key
student ={"name":"talha","marks":88,
          "city":"Karachi","age":20}
student.pop("age")
student
```

```
{'name': 'talha', 'marks': 88, 'city': 'Karachi'}
```

```
#keys,values,items (json file)
student ={"name":"atif","marks":98,
          "city":"Lahore","age":29}
student.keys()
```

```
dict_keys(['name', 'marks', 'city', 'age'])
```

```
student.values()
```

```
dict_values(['atif', 98, 'Lahore', 29])
```

```
student.items()
```

```
dict_items([('name', 'atif'), ('marks', 98), ('city', 'Lahore'), ('age', 29)])
```

```
print(student)
```

```
{'name': 'atif', 'marks': 98, 'city': 'Lahore', 'age': 29}
```

```
type(student)
```

```
dict
```

```
#WAP which take values from the user
account_holders = {}
account_holders["name"] = input("Enter your name: ")
account_holders["age"] = int(input("Enter your age: "))
account_holders["balance"] = float(input("Enter your account balance: "))
print(account_holders)
```

```
Enter your name: uzair
Enter your age: 55
```

```
Enter your account balance: 67
{'name': 'uzair', 'age': 55, 'balance': 67.0}
```

```
#only access by keys
candidate = {
"name": "Ayesha",
"score": {
"python": 88,
"database": 91,
"ai": 86
}
}
print(candidate["score"]["ai"])
```

```
86
```

```
#access dictionary values by key
candidate = {
"name": "Ayesha",
"score": {
"python": [12,32],
"database": [23,45],
"ai": [25,35]
}
}
print(candidate["score"]["ai"])
```

```
[25, 35]
```

```
#access values by key and list index
candidate = {
"name": ["Ayesha","ali"],
"score": {
"python": [12,32],
"database": [23,45],
"ai": [25,35]
}
}
print(candidate["score"]["ai"][0])
```

```
25
```

```
print(candidate["name"])
```

```
Ayesha
```

```
applicant = {"name": "Ali", "age": 21}
print(applicant.get("name"))
```

```
Ali
```

```
applicant["name"]
```

```
'Ali'
```

```
#sets
numbers = {10,20,30}
print(numbers)
```

```
{10, 20, 30}
```

```
type(numbers)
```

```
set
```

```
empty = {0}
type(empty)
```

```
set
```

```
empty_2 = ()
type(empty_2)
```

```
tuple
```

```
a = set()
print(a)
type(a)
```

```
set()
set
```

```
b = {""}
type(b)
```

```
set
```

```
colors = {"red","yellow","green"}
colors.add("orange")
colors
```

```
{'green', 'orange', 'red', 'yellow'}
```

```
numbers = {1,2,3,1,4,3}
numbers
```

```
{1, 2, 3, 4}
```

```
colors = {"red","yellow","green"}
colors.remove("red")
colors
```

```
{'green', 'yellow'}
```

```
s1 = {1,2,3,4}
s2 = {3,4,5,6,7}
print(s1.union(s2))
```

```
{1, 2, 3, 4, 5, 6, 7}
```

```
s1 = {1,2,3,4}
s2 = {3,4,5,6,7}
s1.intersection(s2)
```

```
{3, 4}
```

```
s1 = {1,2,3,4}
s2 = {3,4,5,6,7}
s1.difference(s2)
```

```
{1, 2}
```

```
#Loops
#for iterator(variable) in range/list/tuple:
  #statement
```

```
num = [1,2,3,4,5,6]
for i in num:
  print(i)
```

```
1
2
3
4
5
6
```

```
num = [1,2,3,4,5,6]
for i in num:
  print(i)
print("Program end")
```

```
1
2
3
4
5
6
Program end
```

```
num = [1,2,3,4,5,6]
for i in num:
  print(i)
  print("Program end")
```

```
1
Program end
2
Program end
3
Program end
```

```
4
Program end
5
Program end
6
Program end
```

```
ch = "Python"
for c in ch: #iterator
  print(c)
```

```
P
y
t
h
o
n
```

```
for n in range(10):
  print(n)
```

```
0
1
2
3
4
5
6
7
8
9
```

```
for n in range(1,10):
  print(n)
```

```
1
2
3
4
5
6
7
8
9
```

```
#WAP to print even number from 2 to 20
for i in range(2,21,2):
  print(i)
```

```
2
4
6
8
10
12
14
16
18
20
```

```
for i in range(2,21,1):
  print(i)
```

```
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

```
#WAP which takes input then print multiplication table
number = int(input("Enter a number: "))
for n in range(1,13):
  print(number,"x",n,"=",number*n)#number x repeat = times
```

```
Enter a number: 11
11 x 1 = 11
11 x 2 = 22
11 x 3 = 33
11 x 4 = 44
11 x 5 = 55
11 x 6 = 66
11 x 7 = 77
11 x 8 = 88
11 x 9 = 99
11 x 10 = 110
11 x 11 = 121
11 x 12 = 132
```

```
#calculate sum from 1 to 10 #0+1=1+2=3+3=6+4=10+5=15+6=21....
#1+2+3+4+5+6+7+8+9+10
total = 0
for i in range(1,11):
  total = total + i #total variable updated after every iteration and number is also added in total
print("Total", total)
```

```
Total 55
```

```
total = 0
for i in range(1,100):
  total += i #total variable updated after every iteration and number is also added in total
print("Total", total)
```

```
Total 4950
```

```
pateint = ["ali","muhammad","maaz","kahlid","anas"]
for p in pateint:
  print(p,end=" ")
```

```
ali muhammad maaz kahlid anas
```

```
pateint = ["ali","muhammad","maaz","kahlid","anas"]
for p in pateint:
  print(p,end=" ")
```

```
ali
muhammad
maaz
kahlid
anas
```

```
#break = to stop the loop
for i in range(1,11):
  if i == 5:
    break
  print(i)
```

```
1
2
3
4
```

```
employee = ["ali","uzair","talha"]
for name in employee:
  print(name)
```

```
ali
uzair
talha
```

```
Start coding or generate with AI.
```